

Examiner
only

6. (a) (i) A car is travelling at initial velocity, u , of 15 m/s when it **decelerates** at a constant rate of 3.5 m/s² for a distance, x , of 20 m.

Use an equation from page 2 to determine the final velocity, v , of the car. [3]

$v = \dots\dots\dots$ m/s

- (ii) During the deceleration, the work done on the driver reduces his kinetic energy from 5 625 J to 2 125 J over a distance of 20 m.

Use the equation:

$$\text{work done} = \text{force} \times \text{distance}$$

to determine the mean force acting on the driver. [3]

Mean force = $\dots\dots\dots$ N

- (b) Airbags are designed to rapidly inflate in the event of a collision. Explain how they help to keep passengers safe. [3]

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